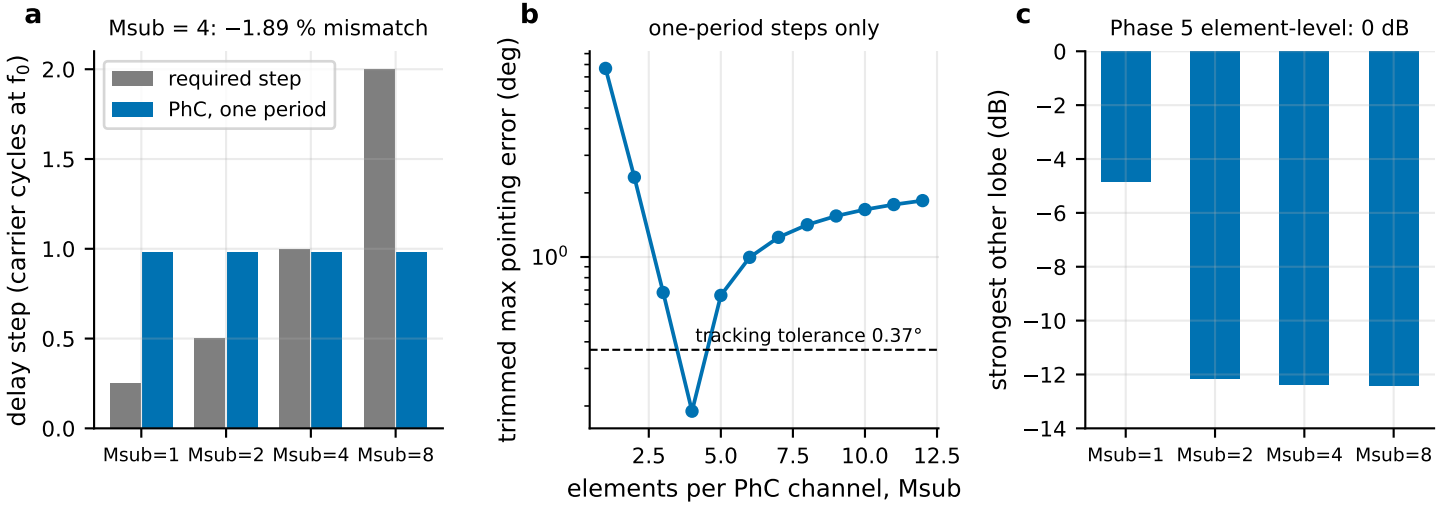
a
Msub = 4: −1.89 % mismatch
required step
PhC, one period
delay step (carrier cycles at f_0)
2.0
1.5
1.0
0.5
0.0
Msub=1
Msub=2
Msub=4
Msub=8
b
one-period steps only
trimmed max pointing error (deg)
10^0
tracking tolerance 0.37°
2.5
5.0
7.5
10.0
12.5
elements per PhC channel, Msub
c
Phase 5 element-level: 0 dB
strongest other lobe (dB)
0
−2
−4
−6
−8
−10
−12
−14
Msub=1
Msub=2
Msub=4
Msub=8